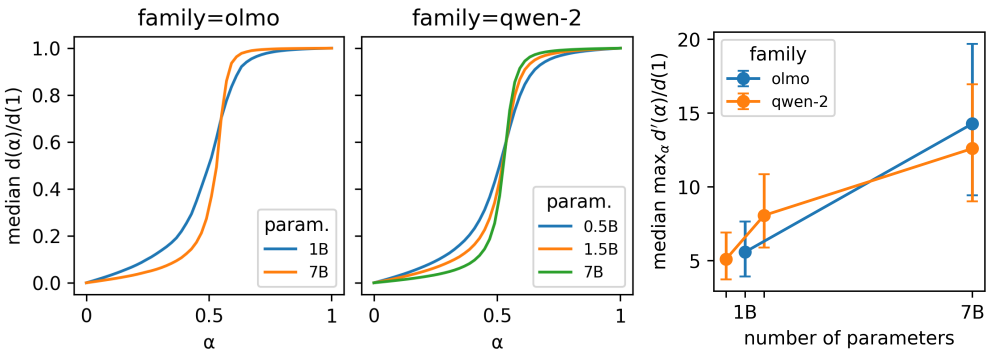


(a) (b) (c)

Figure 3: (a,b) Median relative output distance as a function of interpolation coefficient α for different models from the (a) `OLMo` and (b) `Qwen2` families. (c) Maximum slope as a function of the number of parameters for both model families. Dots represent median, and error bars represent 25th and 75th percentiles.

3. Stable regions are large compared to the polytopes described by Black et al. [2]. When interpolating between semantically different prompts, we appear to cross only a single stable region boundary, while hundreds or thousands of gates change their activation sign during the same interpolation (Appendix F).

Additionally, in Figure 2c, we plot the logit difference, normalized to 0-1 range, between top predictions for both prompts in D1-D3 pairs. The change in logit difference follows a similar dynamic to $d(\alpha)$, suggesting that change in $d(\alpha)$ reflects an interpretable change in model predictions.

Impact of model size We want to test if the results we report in the previous section generalize to other prompts and models, and whether the model size plays a role. We sample 1,000 pairs of 10-token-long, unrelated prompts (p_A, p_B) from the `sedthh/gutenberg_english` [14] dataset[4], and we use the methodology described for the previous experiment.

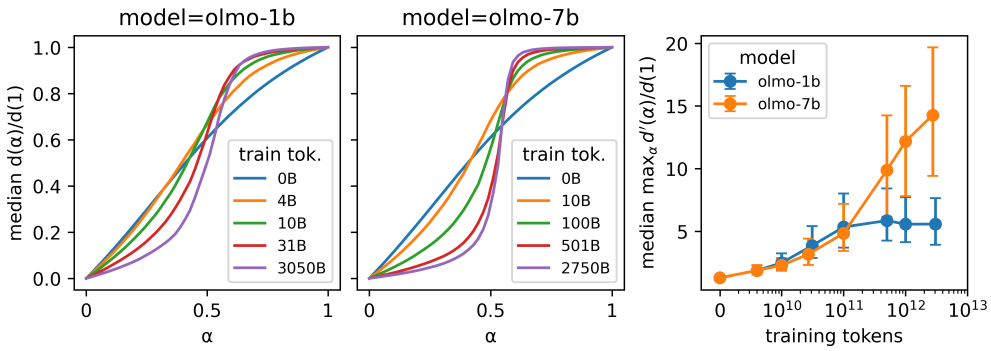
In Figures 3a and 3b we plot the median relative distance for different model sizes from `OLMo` and `Qwen2` families respectively. The curves appear to get sharper with increased model size. We quantify this sharpening effect in Figure 3c by plotting the median of maximum slope of the relative distance, which confirms the qualitative observation.

Impact of training progress We repeat the experiments from the previous section, this time varying the number of training tokens instead of model size. Results presented in Figure 4 reveal several insights. First, curves for randomly initialized models appear close to linear. However, as training progresses, we observe an increasing sharpness in the curves, similar to the effect seen with larger model sizes. This suggests the emergence and refinement of stable regions throughout the training process. Interestingly, the rate of this sharpening effect appears to plateau earlier for smaller `OLMo-1B` model compared to the larger `OLMo-7B`.

At least two different factors may contribute to the sharpening: (1) models develop additional stable regions; (2) the boundaries between existing regions become more defined.[5] To investigate the role of these factors, we examine a 2D slice of the residual stream after the first layer in Figure 1. This slice is spanned by three model-generated activations (A, B, C), corresponding to the red, green, and blue circles respectively. We create synthetic activations within this slice. For each synthetic activation, we compute how similar the model's output is to the outputs produced by A, B, and C. The

[4]We chose this popular dataset of English books, as we expect it was used in training of most Transformer LMs. Additionally, we wanted to have a dataset of a limited diversity, and easily understandable next token predictions, but this will only become relevant in future work.

[5]Factor 1 would result in more individual curves being sharp rather than linear, so the aggregated result would appear sharper. Another factor could be that the location of sharp jump is initially spread for different curves, but becomes more concentrated. That would affect the qualitative plots, (a,b) but not the quantitative plot (c).



(a) (b) (c)

Figure 4: (a,b) Median relative output distance as a function of α for (a) `OLMo-1B` and (b) `OLMo-7B` models. (c) Maximum slope as a function of the number of training tokens for both models. Dots represent median, and error bars represent 25th and 75th percentiles. Note that in subfigures (a) and (b) we only show a few selected checkpoints for readability.

RGB color of each point in the plot represents these similarities: red, green, and blue components correspond to similarities with A, B, and C respectively. See Appendix C for details.

Solid colors indicate stable regions, while color transitions represent boundaries between them. Initially, the plot shows a smooth gradient of colors. As training progresses, distinct regions of solid color emerge, and the boundaries between these regions become increasingly sharp. In Appendix D, we present more slices for different triples of model-generated activations. Some of them hint at existing stable regions splitting during training, in particular Figure 11.

5 Discussion

We only look at a handful of relatively small Transformer language models. We have seen similar results for GPT, Pythia, Phi, and Llama, with a notable exception of Gemma.

We only present indirect evidence for the relatively large size of stable regions, compared to polytopes. Estimating their size or number directly, and how it changes during training and with model size, could provide a more solid ground for this claim.

6 Acknowledgements

This research was conducted as part of the LASR Labs program. Jacek Karwowski was supported by a grant from Open Philanthropy. We would like to thank Lawrence Chan, Daniel Tan, Charlie Griffin, Jake Mendel, Nix Goldowsky-Dill, Bilal Chughtai, and James Wilken-Smith for feedback, and Erin Robertson for guidance.

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